Measuring Linguistic Distance Using Spectral Overlap

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Motivation

Quantifying how *dissimilar* two languages are helps to

- allocate teaching hours efficiently
- guide model-transfer paths in low-resource NLP
- reconstruct family trees in historical linguistics
- prioritise endangered-language support

Gap in Existing Metrics

- Producing distance matrices is **slow, uneven, or narrow in scope**.
- Only *some* typological axes get measured, *some* languages get compared.
- Often requires **costly, labour-intensive fieldwork**.

Opportunity: Language Models and Frequency Analysis

- **Language models**
  - Multilingual transformers internalise phonology, morphology, syntax, semantics.
  - Every token yields **high-dimensional embedding / likelihood**
- **Text signals** : A sequence of embeddings or log-probs can be treated as a *time-series*
- **Frequency analysis looks past words**
  - The Fourier transform highlights **periodicities**
  - It ignores surface token identity and instead inspects **how** information is rhythmically distributed.

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Related Work – Linguistic Distance Metrics

Approach	Signal	Strengths	Blind spots
Cognate / string (Swadesh LD, ASJP)	Word lists	Simple, interpretable	Ignores syntax & discourse
***n*-gram divergences** (KL on trigram LMs)	Probability dists.	Captures ordering	Data sparseness, tokenization bias
Embedding geometry (multilingual skip-gram, mBERT vecs)	Vector space	Dense, model-led	Model-specific; opaque
Typological overlap (WALS, URIEL)	Expert features	Human-readable	Sparse, coarse, costly

*Most compute **time-domain** distances; periodic structure remains untapped.*

Related Work – Signal-Processing in NLP

Text as a signal

- Spectral peaks in letter / word streams → authorship, topic keywords.
- DFT reveals **long-range correlations** and stylistic rhythms.

Neural frequency signatures

- Embedding eigen-spectra expose anisotropy.
- Attention heads approximate Kalman filters.

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Research Questions

1. **RQ1** – Do spectral metrics correlate with phonemic, lexical, cognitive & pedagogical baselines?
2. **RQ2** – Are the metrics model- and dataset-agnostic?
3. **RQ3** – Which frequency bands drive the correlation, and do they reflect typological traits?

Main Contributions

- Introduce **Spectral KL** & **Spectral Coherence** distances
- Evaluate 100 languages, 2 transformers, 2 corpora, 3 metrics
- Map discriminative frequency bands to linguistic features

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Background: Linguistic Subsystems

Subsystem	Captures	Typical distance
Phonetics	Raw sounds	IPA edit distance
Phonology	Sound system rules	Jaccard inventories
Morphology	Word structure	Morphemes/word
Syntax	Phrase order	Tree-edit
Semantics	Meaning	Embedding cosine

Baselines

- **Phonemic** – ASJP LDND
- **Lexical** – Swadesh Levenshtein
- **Mutual intelligibility** – cloze tests
- **FSI learning time** – weeks for English L1 diplomats

Language Modelling in a Nutshell

Goal: assign a probability to a token sequence

$P(w_{1:n}) = \prod_{t=1}^{n} P(w_t \mid w_{1:t-1})$

Model	Context window	Drawbacks
n-gram, Katz/Kneser smoothing	fixed (n−1) tokens	sparsity, short range
Continuous *n*-gram (word2vec)	fixed	still local
RNN / LSTM / GRU	*unbounded* (sequential)	slow, vanishing grads
Transformers	full sentence (parallel)	–

Neural Language Models

Recurrent (LSTM/GRU)

$$\mathbf{h}_t = f(\mathbf{h}_{t-1}, \mathbf{e}(w_t))$$

unbounded history, but sequential updates.

Attention replaces recurrence

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\big(QK^\top / \sqrt{d_k}\big)V$$

Transformer layer = Multi-Head Attention + Feed-Forward + Residual & LayerNorm.

Stacking layers yields deeply contextual embeddings

mBERT/XLM-R

Pre-trained language encoders that share parameters across dozens of languages.

- **Training objective** – masked-language modelling (MLM) on mixed-language corpora
- **Sub-word vocabulary** – WordPiece (mBERT 110 k) or SentencePiece-BPE (XLM-R 250 k) built jointly across languages → **shared embedding space**.
- **Emergent alignment**
    - Cross-lingual tokens with similar context land near one another.
    - Mid-layers capture phonology & morphology; upper layers encode syntax & semantics.

Fourier Transform

Transforms a time-domain signal into its frequency components.

$$X(f) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi f t}\, dt$$

- **Discrete variant**:

  $$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j\,2\pi k n/N}, \qquad k = 0, \ldots, N-1$$

- **Energy <> variance**:

  $$\sum_n |x[n]|^2 = \frac{1}{N} \sum_k |X[k]|^2$$

Welchs Method

Variance-reduced estimate of the power spectral density (PSD).

1. **Segment** the signal into (K) overlapping frames of length (L) (e.g. 50 % overlap, Hann window (w[n])).
2. **Window** each frame: $x_i^w[n] = x_i[n]\ w[n]$.
3. **Periodogram per frame**

   $$P_i(f) = \tfrac{1}{U}\left|\mathcal{F}\{x_i^w\}(f)\right|^2, \quad U = \tfrac{1}{L}\textstyle\sum_n w^2[n]$$

4. **Average** the periodograms

   $$\hat{P}_{\text{Welch}}(f) = \tfrac{1}{K}\textstyle\sum_{i=1}^{K} P_i(f)$$

Why Spectra?

- Language encodes **periodicities** (stress, affixes, phrase lengths)
- Comparing PSDs asks *“How is information rhythmically distributed?”*
- Captures structure missed by token-level overlap

Periodicities in Text

Device	Scale	Periodicity pattern	Example
Ablaut reduplication	Token	Fixed vowel alternation (e.g., *i–a–o*) inside paired morphemes	*ping-pong*, *tic-tac-toe*
Anadiplosis	Word	End-of-clause word becomes start of next clause	Fear leads to *anger*; *anger* leads to *hate*; *hate* leads to suffering.
Epistrophe	Phrase	Identical phrase closes successive units	…of the *people*, by the *people*, for the *people*.
Antithetical parallel	Sentence	Mirrored clauses with inverse relation	Ask not what your country can do for you; ask what you can do for your country.

Coherence

Coherence measures the degree to which one signal $x(t)$ predicts another $y(t)$ via an optimal least-squares estimator.

$$C_{xy}(f) = \frac{|G_{xy}(f)|^2}{G_{xx}(f)\,G_{yy}(f)}$$

$$G_{xy}(f) = \mathcal{F}\{R_{xy}(\tau)\} = \int_{-\infty}^{\infty} R_{xy}(\tau)\,e^{-j2\pi f\tau}\,d\tau = \mathbb{E}\{X(f)\,Y^*(f)\}$$

$R_{xy}(\tau) = \mathbb{E}\{x(t)\,y^*(t+\tau)\}$ is the cross-correlation function

Coherence

- **Geometric view:** $|\cos\theta|$ between time-series vectors
- **Range:** $0 \leq C_{xy}(f) \leq 1$
  - **0** $\rightarrow$ **low coherence:** non-linear link, extra inputs, or high noise
  - **1** $\rightarrow$ **high coherence:** strong linear predictability
- **Symmetric:** $|G_{xy}(f)| = |G_{yx}(f)|$
- **Practical use:** frequency-resolved health check
  - High-C_{xy} band $\Rightarrow$ system behaving as modelled
  - Low-C_{xy} band $\Rightarrow$ investigate fault (e.g., lubrication, impurities)

KL Divergence

$$D_{\mathrm{KL}}(P \,\|\, Q) \;=\; \sum_i P(i)\, \log \frac{P(i)}{Q(i)}$$

- **Interpretation:** the surprisal from using Q as a model, when the true probability distribution is P
- **Non-negative:** $D_{\mathrm{KL}} \geq 0$
- **Asymmetric:** $D_{\mathrm{KL}}(P \,\|\, Q) \neq D_{\mathrm{KL}}(Q \,\|\, P)$

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Proposed Metrics

- **Spectral Overlap**

$$PSO(l_1, l_2) = \sum_{f} min(P_1(f), P_2(f))$$

- **Spectral KL Divergence**

$$D_{\mathrm{KL}}\big(l_1 \parallel l_2\big) \;=\; \sum_{f \in \mathcal{F}} P_{l_1}(f)\, \log \frac{P_{l_1}(f)}{P_{l_2}(f)}$$

- **Magnitude-Squared Coherence**

$$D_{\mathrm{C}}\big(l_1, l_2\big) = \frac{1}{K} \sum_{k=1}^{K} \frac{|G_{l_1 l_2}(f_k)|^2}{G_{l_1 l_1}(f_k)\, G_{l_2 l_2}(f_k)}$$

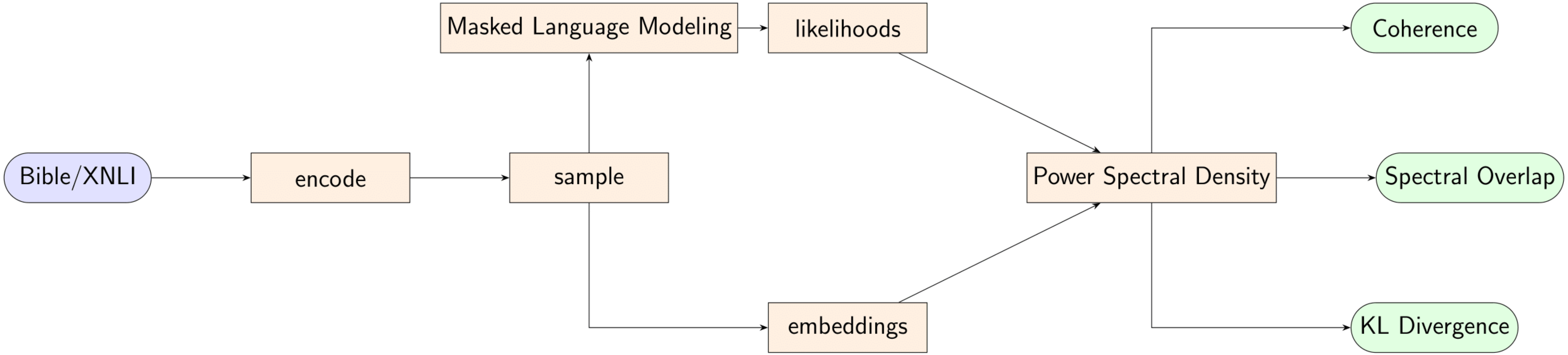
Datasets

Corpus	Domain / unit	Languages	Sampled size	Remarks
XNLI (Conneau 2018)	Multi-genre sentences	15 (14 + EN)	600 × sentences (≥ 20 tokens)	Machine-translated; balanced modern prose
Bible-Corpus (Christodouloupoulos 2015)	Verse-aligned scripture	100	600 × verses (≥ 20 tokens)	39 langs < 1 M speakers; broad family spread

Models

Model	Parameters	Vocabulary size	Languages
mBERT	110 M	110 k	104
XLM-R (base)	270 M	250 k	100

Experimental Pipeline



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Experiments

E1 Time- vs. frequency-domain signals
E2 Model robustness (mBERT / XLM-R)
E3 Dataset robustness (Bible / XNLI)
E4 Layer-wise probe (12 layers)
E5 Sliding freq. windows (.04 & .10 BW)
E6 WALS feature probing

RQ1 – Do spectral metrics correlate with phonemic, lexical, cognitive & pedagogical baselines?
RQ2 – Are the metrics model- and dataset-agnostic?
RQ3 – Which frequency bands drive the correlation, and do they reflect typological traits?

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Key Results

Baseline	Pearson r	Note
Phonemic	**0.62**	--
Lexical	**0.67**	--
Intellig.	**−0.54**	--
FSI weeks	**0.52**	*Spearman

E1: Time vs Frequency (Likelihood)

		Time-Domain Likelihoods				Power Spectra				
baseline	**method**	p_{coef}	p_{pval}	s_{coef}	s_{pval}	p_{coef}	p_{pval}	s_{coef}	s_{pval}	n
mut_int	overlap	−0.001	0.995	0.132	0.353	0.056	0.694	−0.029	0.840	52
	kl	0.184	0.192	0.151	0.287	−0.084	0.553	0.001	0.994	52
lex_sim	overlap	0.060	0.731	0.047	0.790	−0.089	0.612	−0.162	0.353	35
	kl	0.161	0.357	0.259	0.134	0.056	0.748	0.127	0.469	35
fsi	overlap	0.327	0.063	0.250	0.161	**0.416**	0.016*	**0.538**	0.001*	33
	kl	0.195	0.276	0.005	0.978	**0.507**	0.003*	**0.567**	0.001*	33
pho_sim	overlap	0.038	0.433	0.114	0.017*	**0.322**	0.000*	**0.319**	0.000*	435
	kl	0.096	0.045*	0.028	0.561	**0.310**	0.000*	**0.309**	0.000*	435

Likelihood correlations comparing distance metrics in the time domain and frequency domain. *$p < 0.05$. Bold indicates the larger (absolute) coefficient of the two significant values in each time-vs-frequency pair.

E1: Time vs Frequency (Embeddings)

		Time-Domain Embeddings				Power Spectra				
baseline	**distance method**	p_{coef}	p_{pval}	s_{coef}	s_{pval}	p_{coef}	p_{pval}	s_{coef}	s_{pval}	n
mut_int	overlap	**−0.505**	0.000*	−0.527	0.000*	−0.499	0.000*	**−0.550**	0.000*	52
	kl	**−0.499**	0.000*	**−0.534**	0.000*	−0.389	0.004*	−0.403	0.003*	52
lex_sim	overlap	0.568	0.000*	0.564	0.000*	**0.610**	0.000*	**0.618**	0.000*	35
	kl	**0.560**	0.000*	0.555	0.001*	0.484	0.003*	**0.568**	0.000*	35
fsi	overlap	0.146	0.419	0.342	0.052	0.333	0.058	**0.551**	0.001*	33
	kl	0.141	0.435	0.352	0.045*	**0.468**	0.006*	**0.518**	0.002*	33
pho_sim	overlap	0.393	0.000*	0.315	0.000*	**0.584**	0.000*	**0.562**	0.000*	435
	kl	0.380	0.000*	0.311	0.000*	**0.618**	0.000*	**0.547**	0.000*	435

Embedding correlations comparing distance metrics in the time domain and frequency domain. *$p < 0.05$. Bold indicates the larger (absolute) coefficient of the two significant values in each time-vs-frequency pair.

E1: Coherence

metric	Likelihoods				Embeddings				
	p_{coef}	p_{pval}	s_{coef}	s_{pval}	p_{coef}	p_{pval}	s_{coef}	s_{pval}	n
mut_int	−0.371	0.007*	−0.423	0.002*	**−0.537**	0.000*	**−0.543**	0.000*	52
lex_sim	0.394	0.019*	0.342	0.044*	**0.670**	0.000*	**0.661**	0.000*	35
fsi	0.077	0.671	0.267	0.133	0.317	0.072	**0.516**	0.002*	33
pho_sim	0.281	0.000*	0.246	0.000*	**0.623**	0.000*	**0.486**	0.000*	435

Spearman (s_{coef}) and Pearson (p_{coef}) correlations for likelihoods and embeddings with the coherence measure.
*p < 0.05. Bold marks the larger (absolute) significant coefficient in each likelihood-vs-embedding pair.

E2: Metric Robustness to Models (Likelihood)

		mBERT Likelihoods				XLM-R Likelihoods				
baseline	**distance method**	p_{coef}	p_{pval}	s_{coef}	s_{pval}	p_{coef}	p_{pval}	s_{coef}	s_{pval}	n
mut_int	overlap	0.056	0.694	−0.029	0.840	**−0.324**	0.019*	−0.263	0.059	52
	kl	−0.084	0.553	0.001	0.994	−0.233	0.097	−0.221	0.115	
	coherence	**−0.371**	0.007*	**−0.423**	0.002*	−0.360	0.009*	−0.353	0.010*	
lex_sim	overlap	−0.089	0.612	−0.162	0.353	0.188	0.279	0.203	0.243	35
	kl	0.056	0.748	0.127	0.469	0.089	0.610	0.149	0.392	
	coherence	0.394	0.019*	0.342	0.044*	0.460	0.005*	**0.520**	0.001*	
fsi	overlap	0.416	0.016*	0.538	0.001*	**0.492**	0.003*	**0.738**	0.000*	34
	kl	**0.507**	0.003*	0.567	0.001*	0.462	0.006*	**0.723**	0.000*	
	coherence	0.077	0.671	0.267	0.133	**0.399**	0.019*	0.331	0.056	
pho_sim	overlap	0.322	0.000*	0.319	0.000*	**0.360**	0.000*	**0.462**	0.000*	465
	kl	0.310	0.000*	0.309	0.000*	0.346	0.000*	**0.490**	0.000*	
	coherence	**0.281**	0.000*	**0.246**	0.000*	0.234	0.000*	0.145	0.002*	

Likelihood spectral metric correlations with mBERT and XLM-R on the updated dataset. Highlighted cells mark coefficient pairs whose absolute difference exceeds 0.10; within those pairs, the larger significant coefficient is additionally shown in bold. *p < 0.05.

E2: Metric Robustness to Models (Embedding)

		mBERT Embeddings				XLM-R Embeddings				
baseline	**distance method**	p_{coef}	p_{pval}	s_{coef}	s_{pval}	p_{coef}	p_{pval}	s_{coef}	s_{pval}	n
mut_int	overlap	**−0.499**	0.000*	**−0.550**	0.000*	−0.498	0.000*	−0.463	0.001*	52
	kl	−0.389	0.004*	−0.403	0.003*	**−0.528**	0.000*	**−0.520**	0.000*	
	coherence	**−0.537**	0.000*	**−0.543**	0.000*	−0.411	0.002*	−0.399	0.003*	
lex_sim	overlap	**0.610**	0.000*	**0.618**	0.000*	0.499	0.002*	0.597	0.000*	35
	kl	**0.484**	0.003*	**0.568**	0.000*	0.438	0.008*	0.407	0.015*	
	coherence	**0.670**	0.000*	**0.661**	0.000*	0.626	0.000*	0.629	0.000*	
fsi	overlap	0.333	0.058	0.551	0.001*	**0.536**	0.001*	**0.573**	0.000*	33
	kl	0.468	0.006*	0.518	0.002*	**0.589**	0.000*	**0.597**	0.000*	
	coherence	0.317	0.072	**0.516**	0.002*	**0.506**	0.002*	0.514	0.002*	
pho_sim	overlap	**0.584**	0.000*	**0.562**	0.000*	0.481	0.000*	0.421	0.000*	465
	kl	**0.618**	0.000*	**0.547**	0.000*	0.427	0.000*	0.518	0.000*	
	coherence	**0.623**	0.000*	**0.486**	0.000*	0.308	0.000*	0.265	0.000*	

Embedding spectral metric correlations with mBERT and XLM-R. Highlighted cells show coefficient pairs that differ by more than 0.10; within each pair, the larger significant coefficient is additionally shown in bold. *$p < 0.05$.

E3: Metric Robustness to Dataset (Likelihood)

		XNLI Corpus				Bible Corpus				
baseline	**distance method**	p_{coef}	p_{pval}	s_{coef}	s_{pval}	p_{coef}	p_{pval}	s_{coef}	s_{pval}	n_{xnli}/n_{bible}
mut_int	overlap	−0.547	0.453	0.000	1.000	−0.056	0.694	0.029	0.840	4/52
	kl	0.698	0.302	0.400	0.600	−0.084	0.553	0.001	0.994	
	coherence	0.547	0.453	0.105	0.895	**−0.371**	0.007*	**−0.423**	0.002*	
lex_sim	overlap	0.979	0.130	**1.000**	0.000*	0.089	0.612	0.162	0.353	3/35
	kl	0.963	0.175	0.500	0.667	0.056	0.748	0.127	0.469	
	coherence	0.832	0.374	0.500	0.667	**0.394**	0.019*	**0.342**	0.044*	
fsi	overlap	0.446	0.110	**0.611**	0.020*	**0.416**	0.016*	0.538	0.001*	14/33
	kl	0.429	0.126	**0.743**	0.002*	**0.507**	0.003*	0.567	0.001*	
	coherence	−0.063	0.831	−0.440	0.115	0.077	0.671	0.267	0.133	
pho_sim	overlap	0.101	0.419	0.106	0.397	**0.322**	0.000*	**0.319**	0.000*	66/435
	kl	0.054	0.669	−0.096	0.442	**0.310**	0.000*	**0.309**	0.000*	
	coherence	−0.196	0.115	−0.195	0.117	**0.281**	0.000*	**0.246**	0.000*	

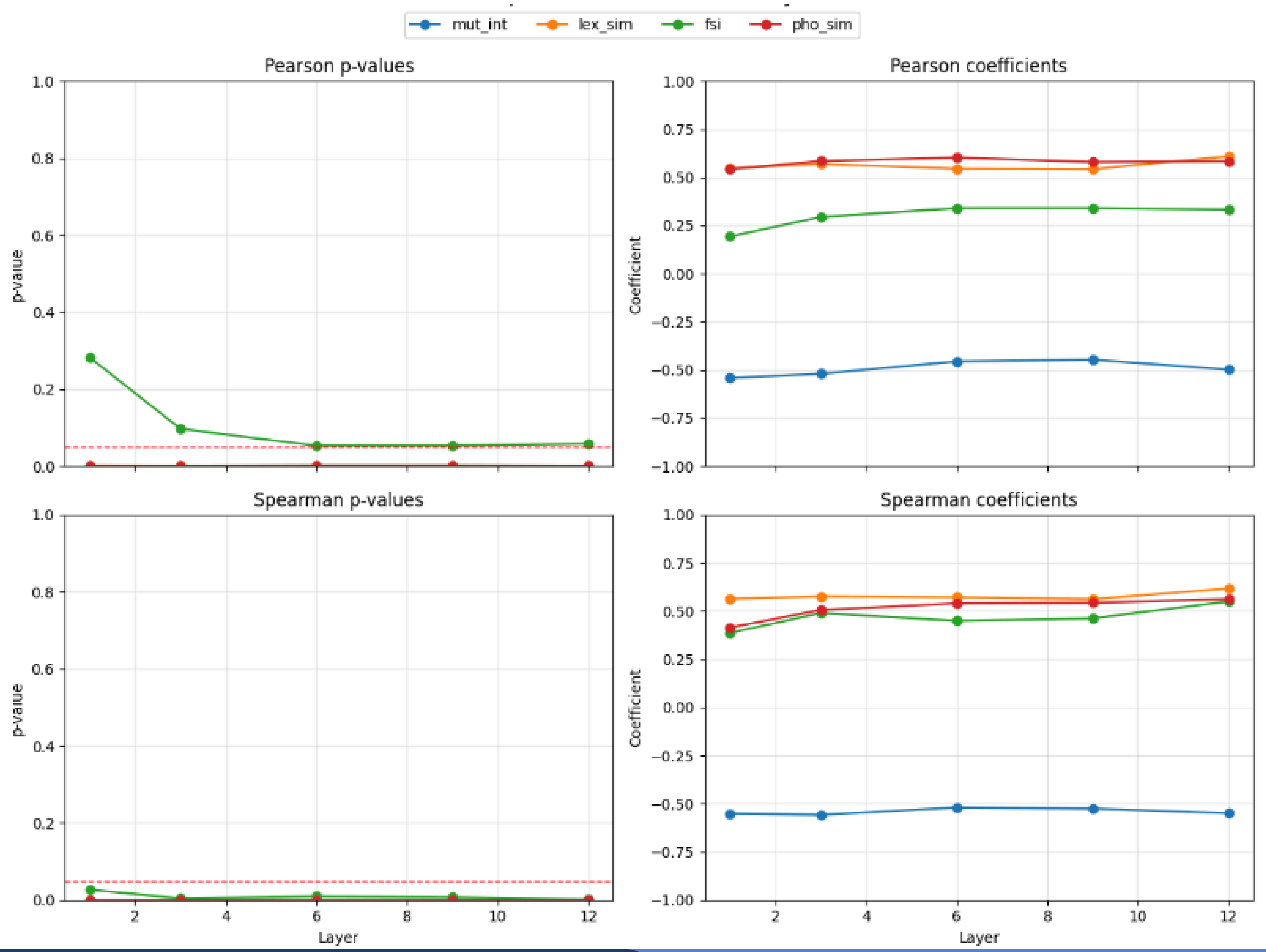
Likelihood spectral metric correlations on the XNLI and Bible corpora. Cells are highlighted when the two coefficients differ by more than 0.10; within those pairs, the larger significant coefficient is additionally shown in bold. *p < 0.05.

		XNLI				Bible Corpus				
baseline	**distance method**	p_{coef}	p_{pval}	s_{coef}	s_{pval}	p_{coef}	p_{pval}	s_{coef}	s_{pval}	n_{xnli}/n_{bible}
mut_int	overlap	−0.547	0.453	0.000	1.000	**−0.499**	0.000*	**−0.550**	0.000*	4/52
	kl	−0.536	0.464	−0.400	0.600	**−0.389**	0.004*	**−0.403**	0.003*	
	coherence	0.547	0.453	0.316	0.684	**−0.537**	0.000*	**−0.543**	0.000*	
lex_sim	overlap	0.954	0.194	0.500	0.667	**0.610**	0.000*	**0.618**	0.000*	3/35
	kl	0.910	0.272	0.500	0.667	**0.484**	0.003*	**0.568**	0.000*	
	coherence	0.251	0.839	0.500	0.667	**0.670**	0.000*	**0.661**	0.000*	
fsi	overlap	0.151	0.606	0.376	0.185	**0.416**	0.016*	**0.538**	0.001*	14/33
	kl	0.351	0.219	**0.582**	0.029*	**0.468**	0.006*	0.518	0.002*	
	coherence	0.177	0.546	0.352	0.217	0.317	0.072	**0.516**	0.002*	
pho_sim	overlap	0.301	0.014*	0.260	0.035*	**0.584**	0.000*	**0.562**	0.000*	66/435
	kl	0.453	0.000*	0.210	0.091	**0.618**	0.000*	**0.547**	0.000*	
	coherence	0.378	0.002*	0.299	0.015*	**0.623**	0.000*	**0.486**	0.000*	

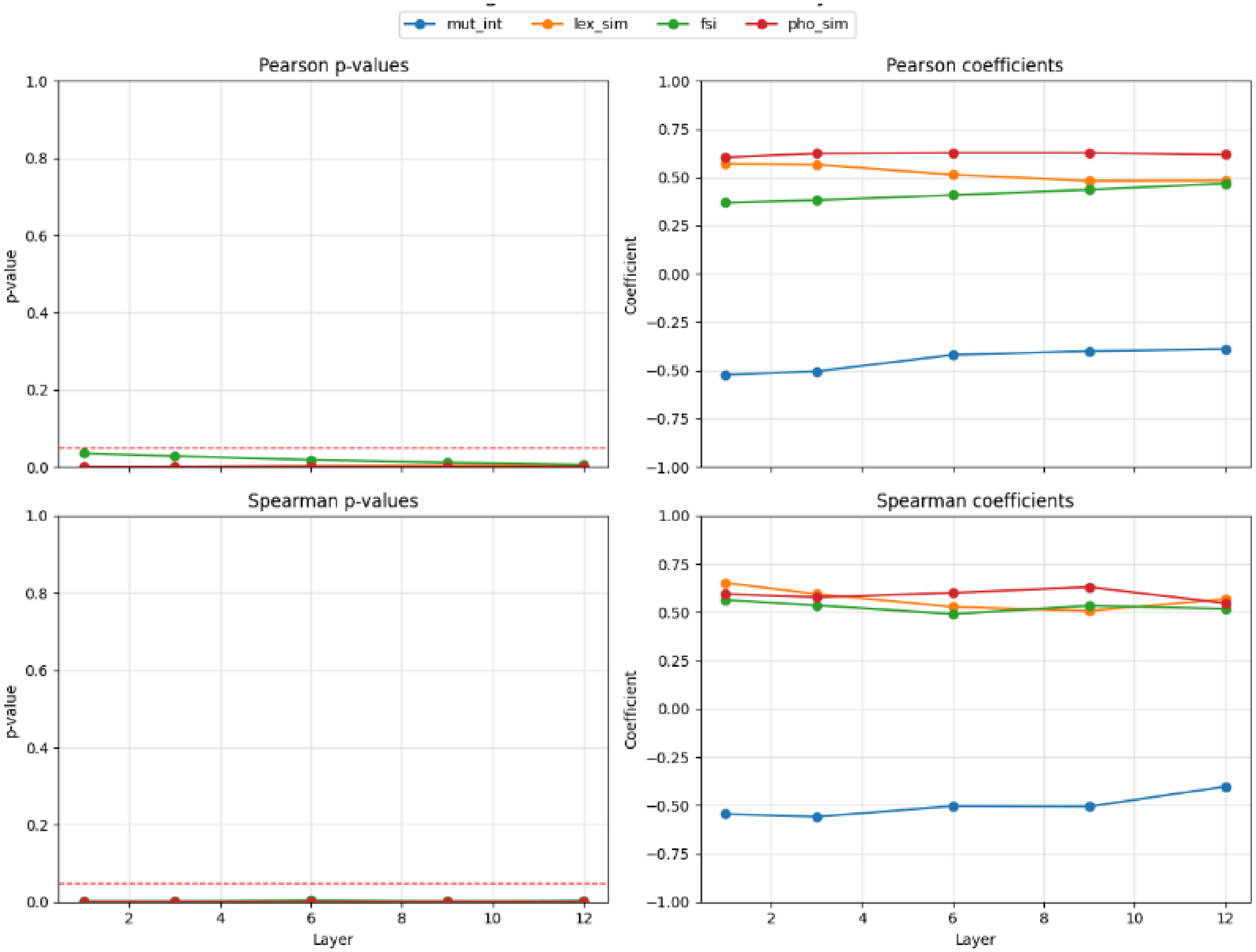
Embedding spectral metric correlations on the XNLI and Bible corpora.

Cells are highlighted when the two coefficients differ by more than 0.10; within those pairs, the larger significant coefficient is additionally shown in bold. *p < 0.05.

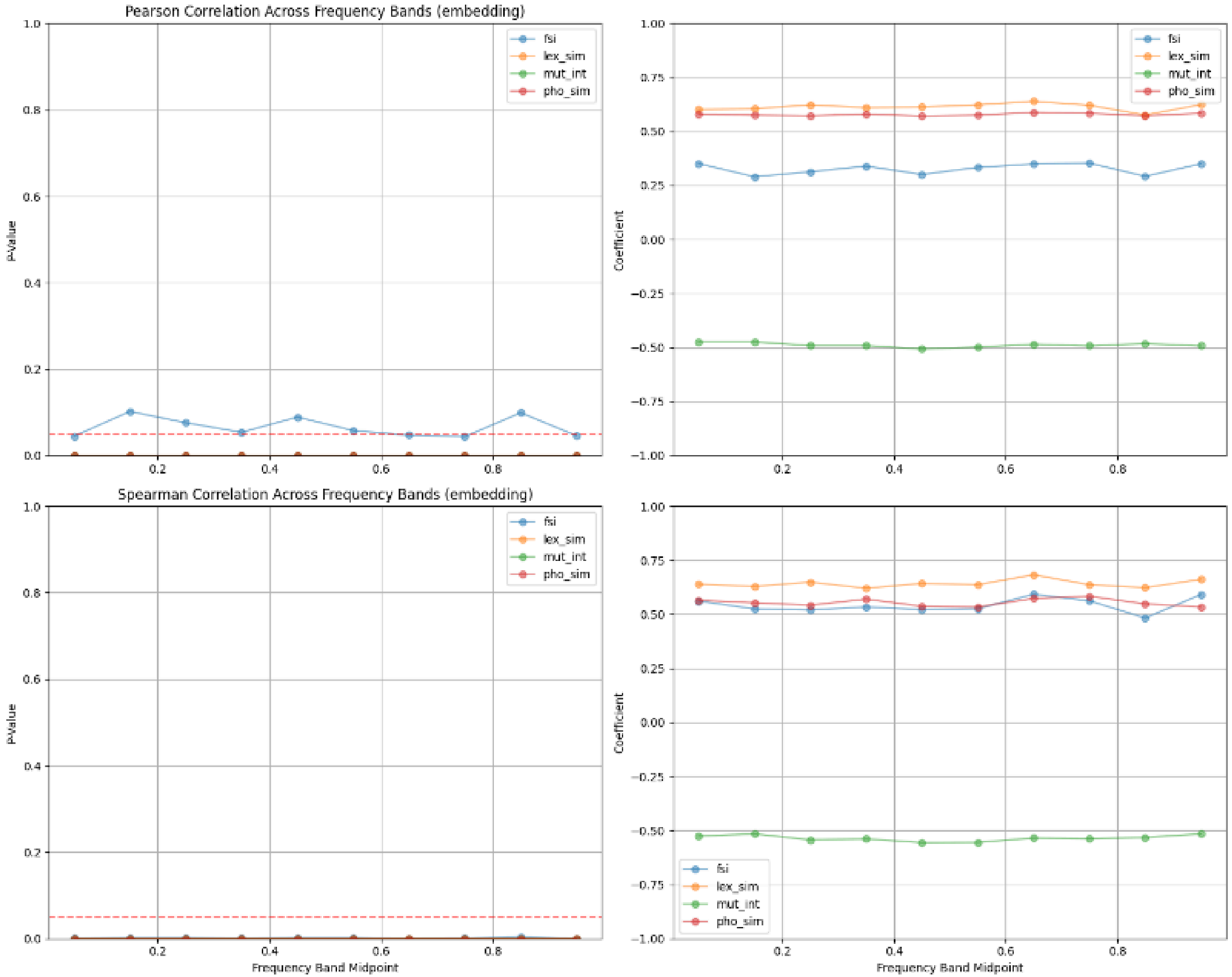
E4: mBERT Layer Investigation (Spectral Overlap)



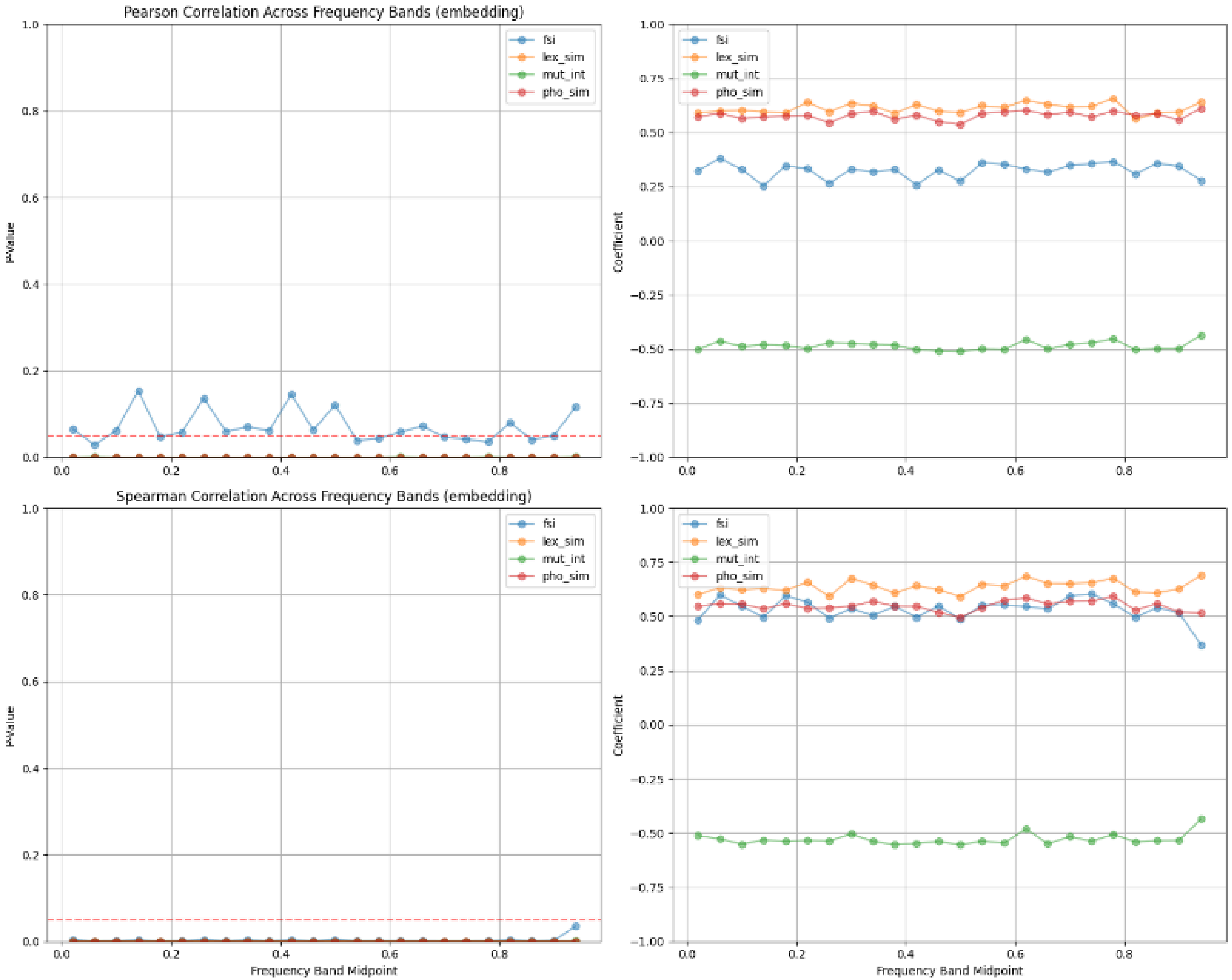
E4: mBERT Layer Investigation (Spectral KL)



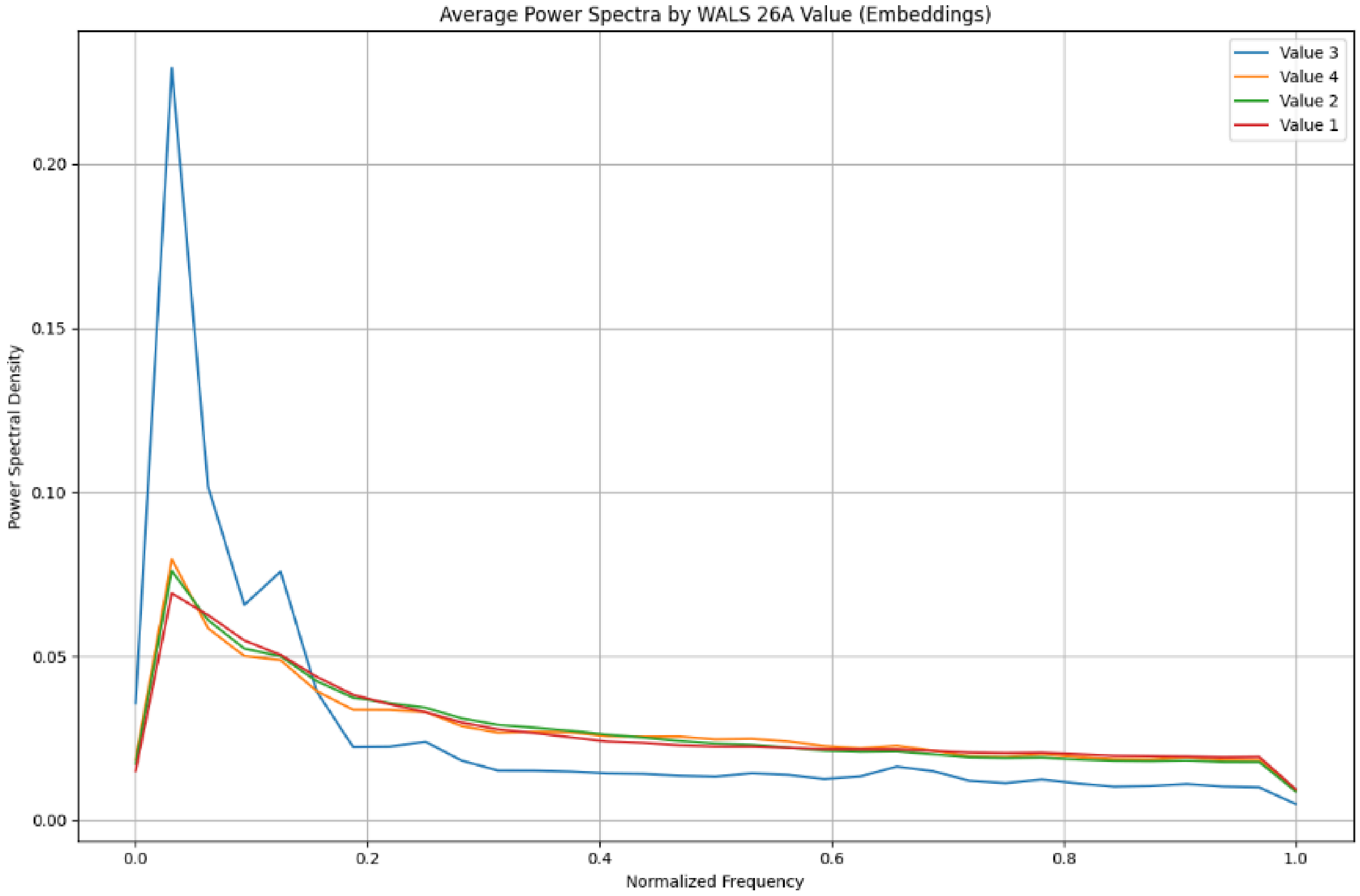
E5: Impact of Band Limiting (Overlap)



E5: Impact of Band Limiting (Overlap)

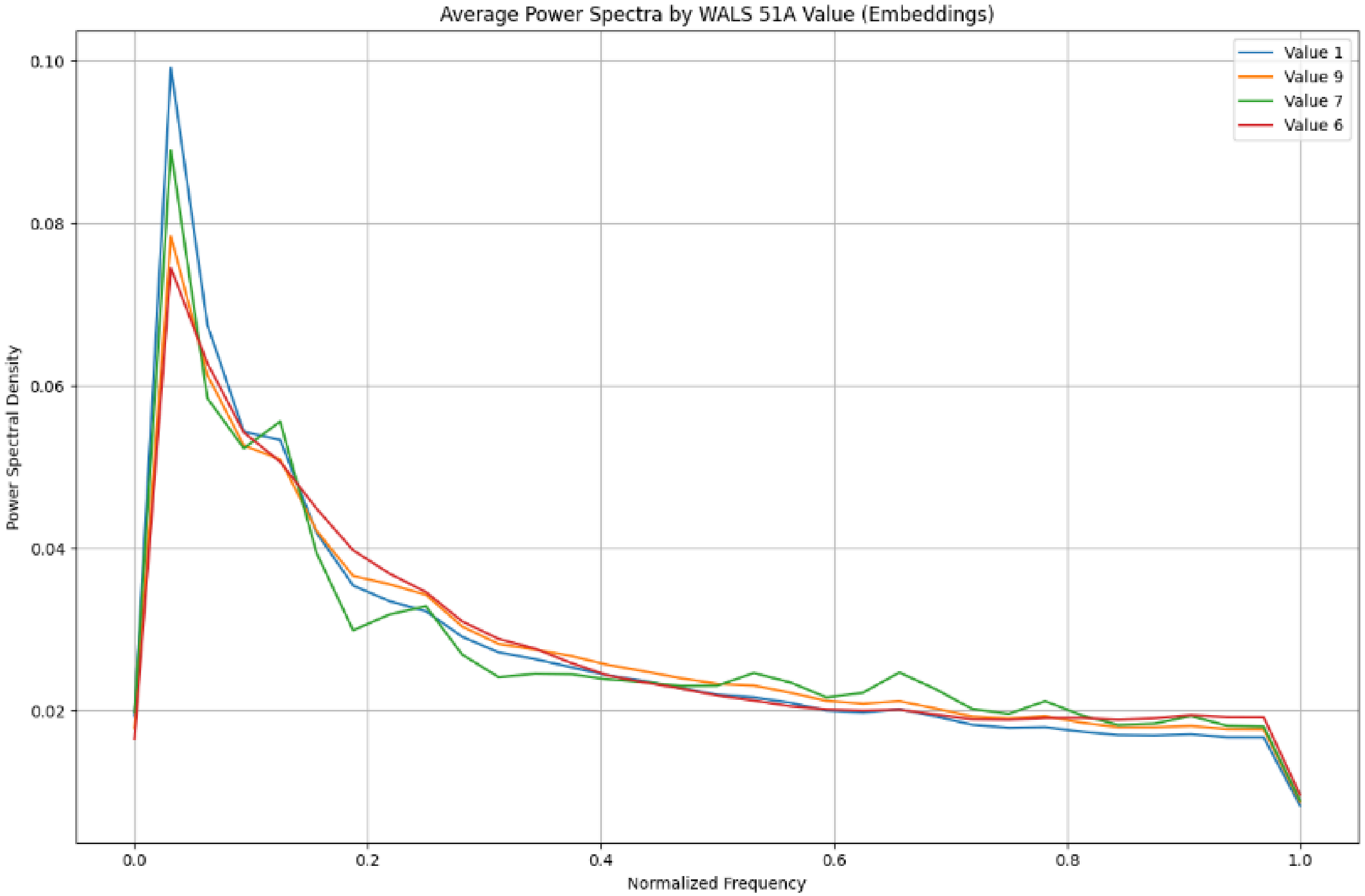


E6: Typological Modulation Effects (26A, Prefixing vs. Suffixing)



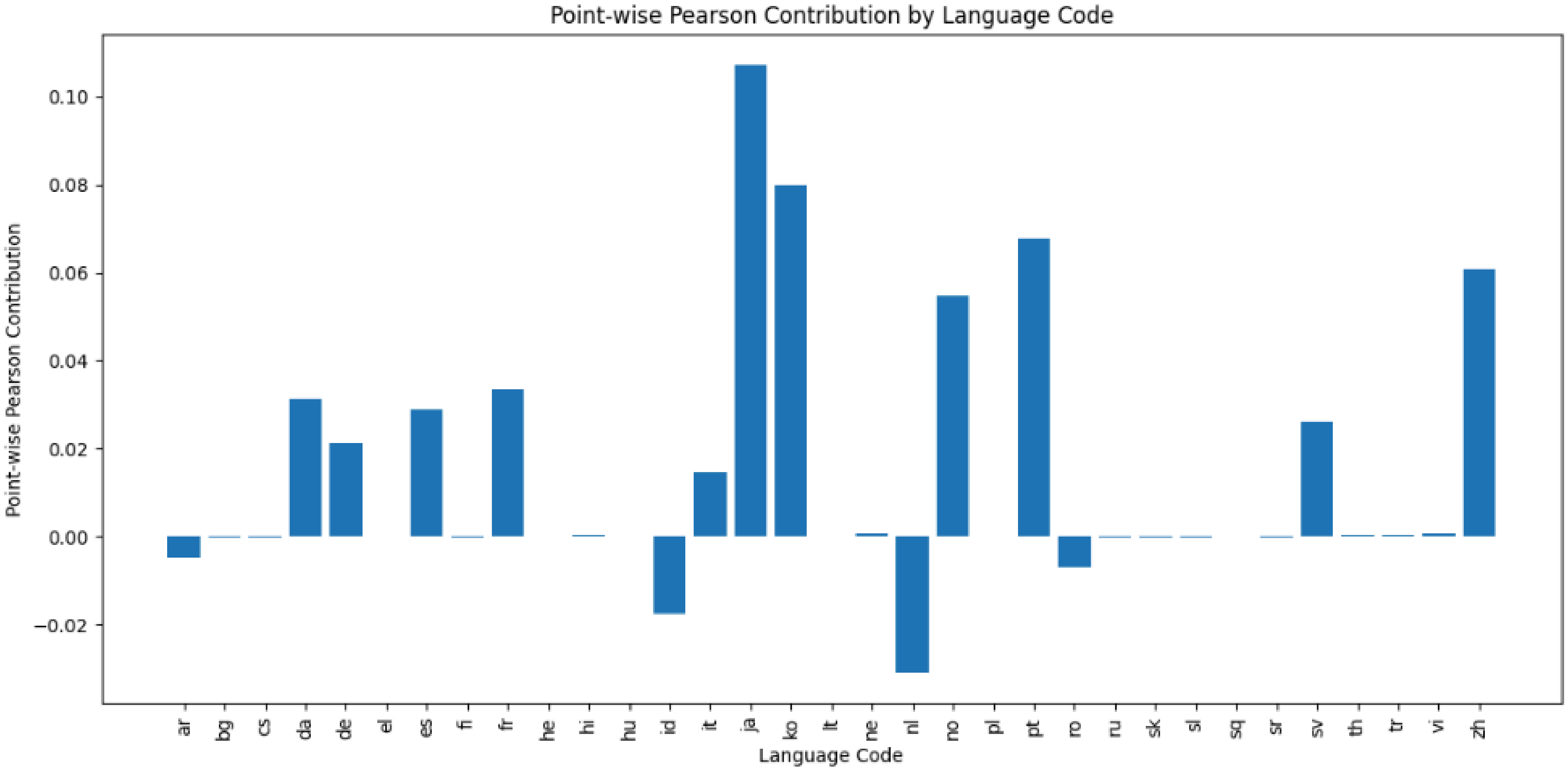
value 3: moderate preference for suffixing

E6: Typological Modulation Effects (51A, Case Affixes)

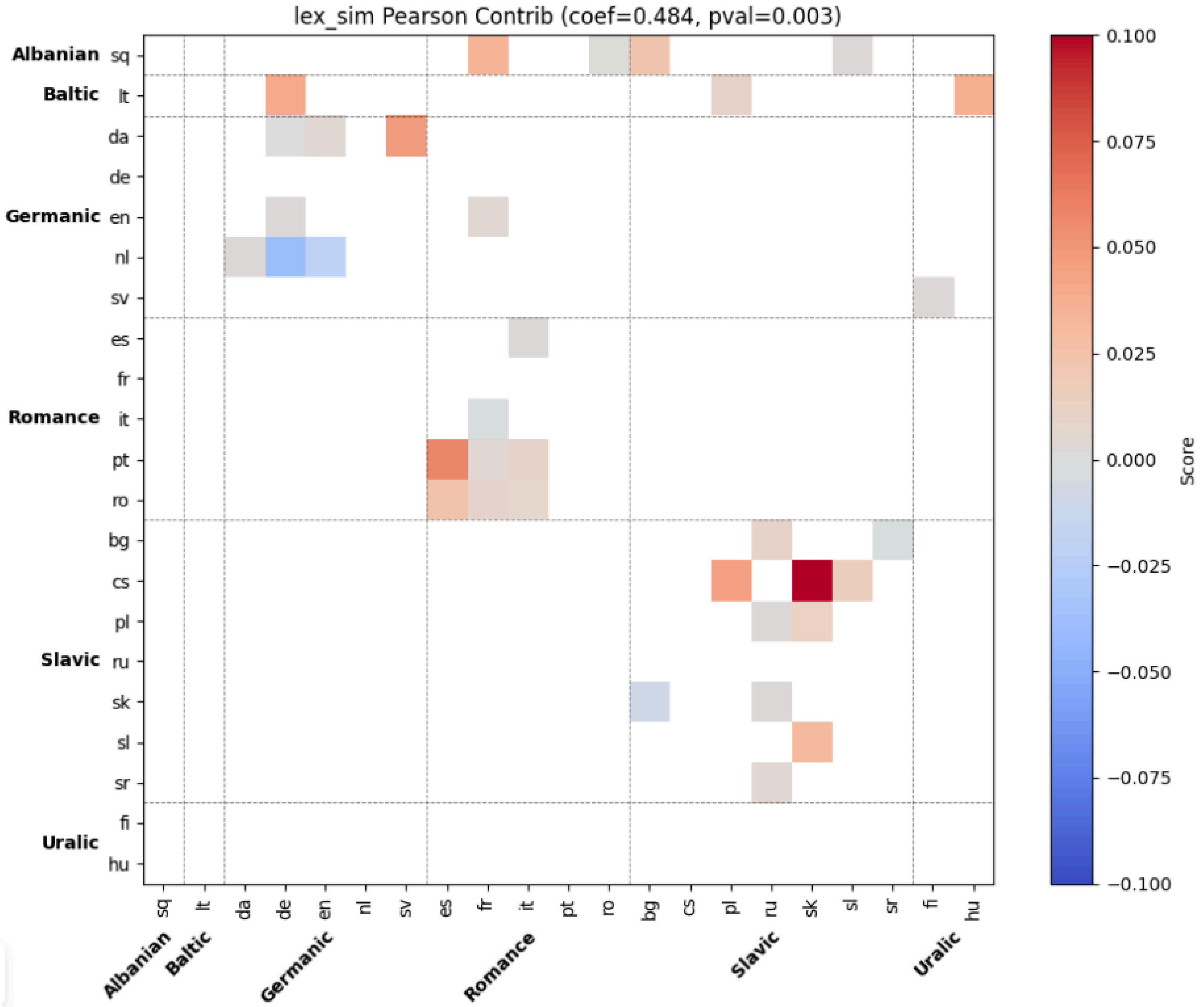


value 9: Neither case affixes nor adpositional clitics

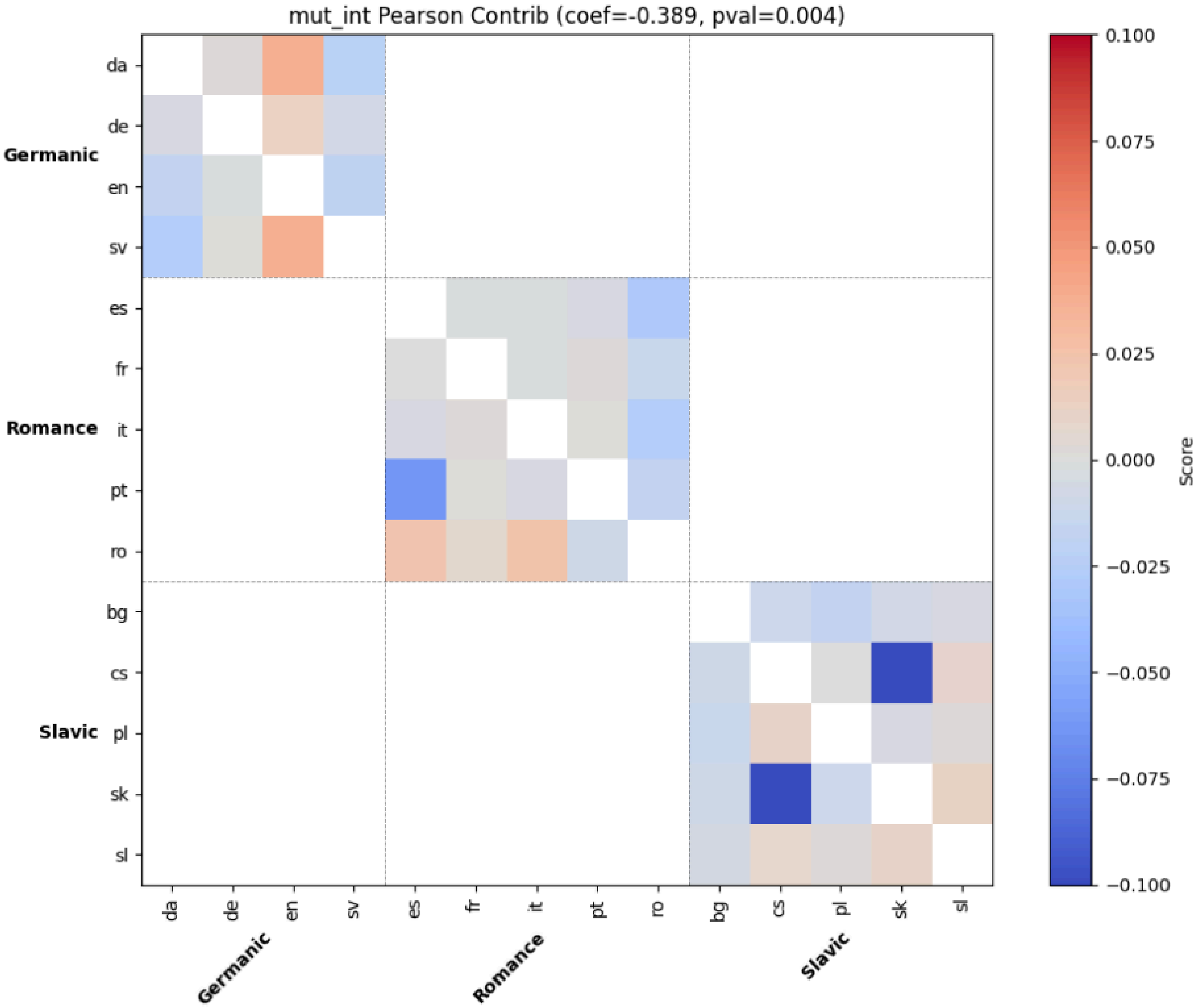
Pointwise Pearson Contribution (Spectral KL, FSI)



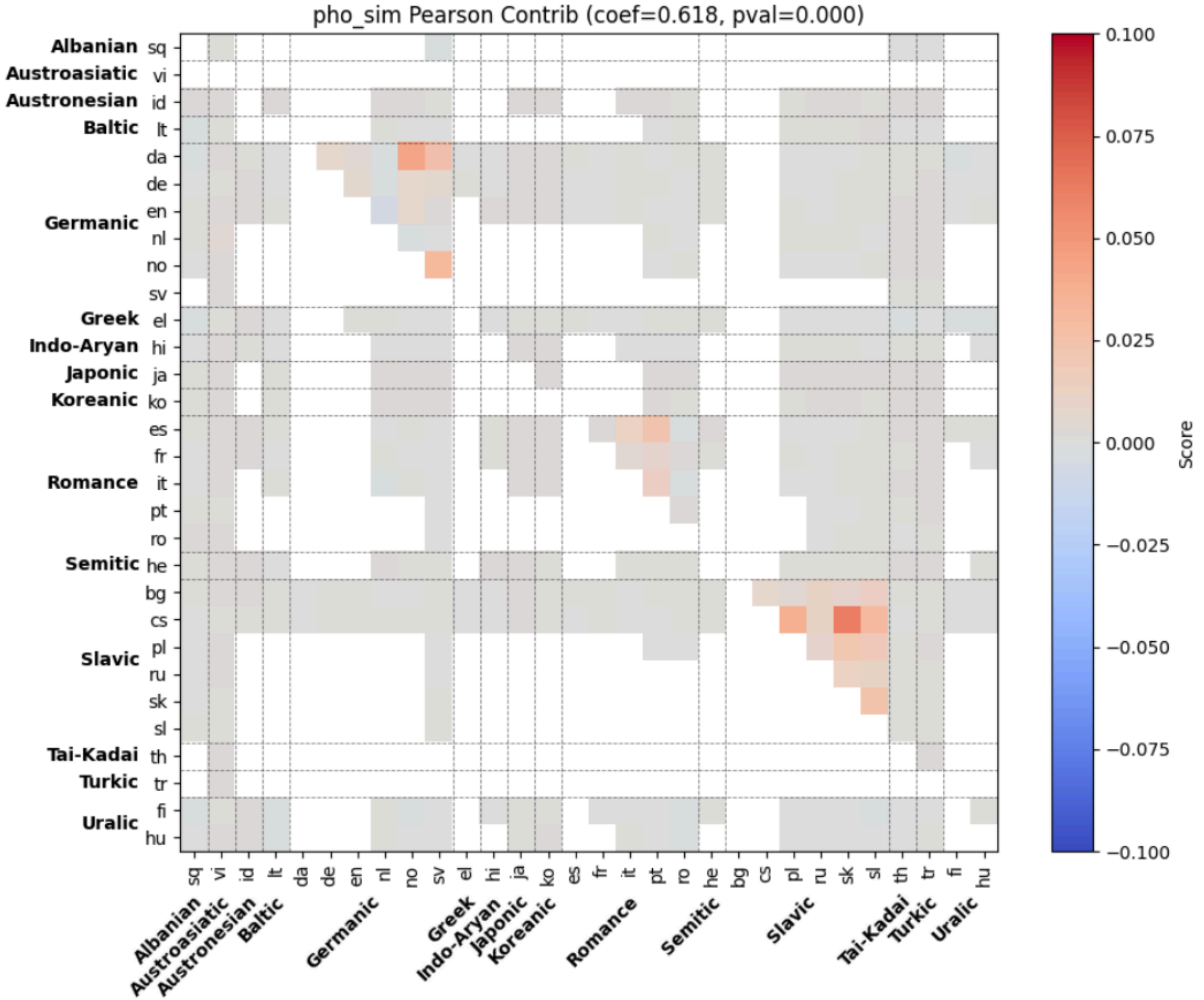
Pointwise Pearson Contribution (Spectral KL, Lexical)



Pointwise Pearson Contribution (Spectral KL, Mutual Intelligibility)



Pointwise Pearson Contribution (Spectral KL, Phonemic)



Wikisize Investigation (Spectral KL)

Baseline	Wikisize Feature	Pearson		Spearman	
		r	p	r	p
mut_int	min_size	0.179	0.205	0.288*	0.039*
	mean_size	0.229	0.103	0.333*	0.016*
	max_size	0.221	0.116	0.321*	0.020*
	diff_size	0.070	0.620	−0.049	0.730
lex_sim	min_size	−0.256	0.138	−0.319	0.062
	mean_size	−0.261	0.131	−0.325	0.057
	max_size	−0.209	0.228	−0.310	0.070
	diff_size	0.056	0.749	0.044	0.801
pho_sim	min_size	0.069	0.152	0.012	0.806
	mean_size	0.043	0.371	−0.040	0.411
	max_size	0.010	0.833	−0.081	0.090
	diff_size	−0.048	0.316	−0.098*	0.041*

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Advantages

- **No linguistic annotation required.**
  Operates on raw text plus a pre-trained multilingual model, bypassing costly typological coding, phonemic transcription, or crowdsourced intelligibility tests.
- **Captures multi-level structure.**
  Transformer embeddings bake in phonological, morphological, and syntactic cues, so the resulting spectra integrate evidence across linguistic subsystems rather than targeting a single facet.
- **Inter-model portability.**
  Comparable results with mBERT and XLM-R suggest the metric is not tied to any specific vocabulary or training regime, making it future-proof as larger multilingual models appear.

Drawbacks

- **Interpretability.**

  While frequency bands may be linked *post hoc* to linguistic phenomena, the mapping is indirect; a spike at (60,%) of Nyquist is harder to interpret than “40 % cognate overlap.”

- **Dependence on subword tokenization.**

  BPE / SentencePiece units blur morpheme boundaries, potentially smearing high-frequency cues such as inflectional affixes or clitics.

- **Asymmetry limits.**

  KL divergence encodes only mild directionality, while overlap and coherence are symmetric. Future work needs metrics whose asymmetry more closely mirrors the sharply asymmetric nature of mutual intelligibility.

Methodological Limitations

The study covers 100 languages, only a fraction of the 7,000+ attested worldwide. Baseline measures are likewise sparse:

- Mutual-intelligibility data exist mainly for closely related languages; the dataset spans just three Indo-European subfamilies.
- The FSI scale is English-centric, rating only distances from English.
- The lexical baseline supports only a subset of language pairs.

Corpus coverage is uneven:

- **Bible** offers 100 languages but a single (often archaic) genre.
- **XNLI** spans just 15 languages, 14 of which are machine-translated from English, though it is at least multi-genre.

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Future Work

- Add syntactic baseline
- Compare time-domain cross-correlation
- Test more genres & dialect continua
- Try **cepstral** analysis for harmonic structures
- test on dialect corpora

Appendix

E3: Metric Robustness to Dataset (Likelihood, XLMR)

		XNLI				Bible Corpus				
baseline	**distance method**	p_{coef}	p_{pval}	s_{coef}	s_{pval}	p_{coef}	p_{pval}	s_{coef}	s_{pval}	n_{xnli}/n_{bible}
mut_int	overlap	−0.547	0.453	0.000	1.000	**−0.324**	0.019*	−0.263	0.059	4/52
	kl	0.653	0.347	0.400	0.600	−0.233	0.097	−0.221	0.115	
	coherence	0.547	0.453	0.000	1.000	**−0.360**	0.009*	**−0.353**	0.010*	
lex_sim	overlap	0.305	0.803	0.500	0.667	0.188	0.279	0.203	0.243	3/35
	kl	−0.782	0.428	**−1.000**	0.000*	0.089	0.610	0.149	0.392	
	coherence	−0.182	0.884	−0.500	0.667	**0.460**	0.005*	**0.520**	0.001*	
fsi	overlap	**0.549**	0.042*	**0.777**	0.001*	0.492	0.003*	0.738	0.000*	14/34
	kl	0.412	0.143	**0.792**	0.001*	**0.462**	0.006*	0.723	0.000*	
	coherence	0.012	0.968	−0.078	0.790	**0.399**	0.019*	0.331	0.056	
pho_sim	overlap	**0.367**	0.002*	0.379	0.002*	0.360	0.000*	**0.462**	0.000*	66/465
	kl	0.292	0.017*	0.340	0.005*	0.346	0.000*	**0.490**	0.000*	
	coherence	−0.105	0.402	−0.062	0.621	**0.234**	0.000*	**0.145**	0.002*	

Likelihood spectral metric correlations on the XNLI and Bible corpora. Cells are highlighted when the two coefficients differ by more than 0.10; within those pairs, the larger significant coefficient is additionally shown in bold. *p < 0.05.

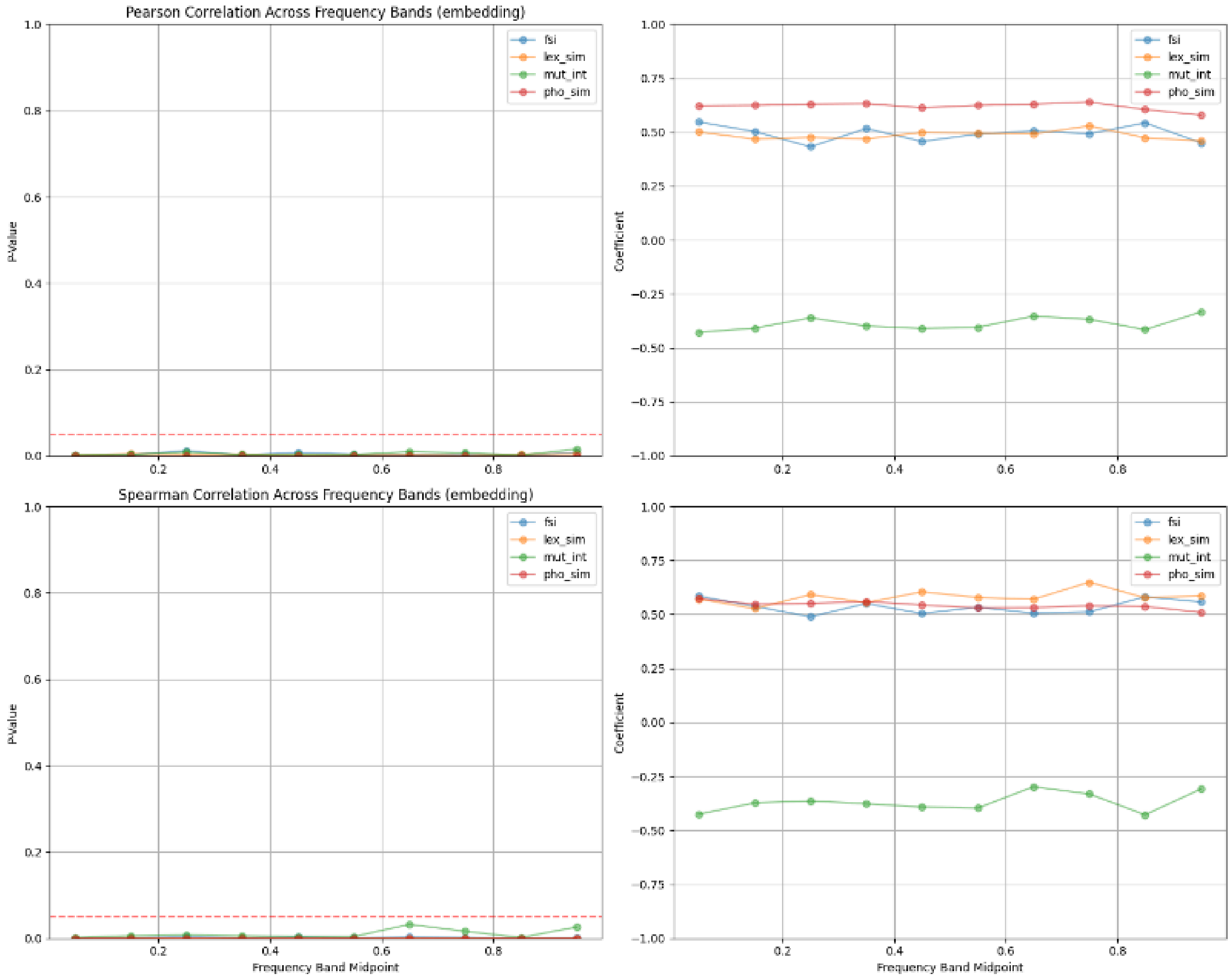
E3: Metric Robustness to Dataset (Embedding, XLMR)

		XNLI				Bible corpus				
baseline	**distance method**	p_{coef}	p_{pval}	s_{coef}	s_{pval}	p_{coef}	p_{pval}	s_{coef}	s_{pval}	n_{xnli}/n_{bible}
mut_int	overlap	−0.547	0.453	0.000	1.000	**−0.498**	0.000*	**−0.463**	0.001*	4/52
	kl	−0.406	0.594	−0.200	0.800	**−0.528**	0.000*	**−0.520**	0.000*	
	coherence	−0.547	0.453	0.000	1.000	**−0.411**	0.002*	**−0.399**	0.003*	
lex_sim	overlap	0.720	0.488	**1.000**	0.000*	**0.499**	0.002*	0.597	0.000*	3/35
	kl	0.811	0.398	**1.000**	0.000*	**0.438**	0.008*	0.407	0.015*	
	coherence	0.598	0.592	0.500	0.667	**0.626**	0.000*	**0.629**	0.000*	
fsi	overlap	0.036	0.903	0.044	0.881	**0.536**	0.001*	**0.573**	0.000*	14/34
	kl	0.290	0.315	0.327	0.253	**0.589**	0.000*	**0.597**	0.000*	
	coherence	−0.140	0.634	−0.152	0.605	**0.506**	0.002*	**0.514**	0.002*	
pho_sim	overlap	0.047	0.706	0.180	0.149	**0.481**	0.000*	**0.421**	0.000*	66/465
	kl	0.221	0.074	0.047	0.711	**0.427**	0.000*	**0.518**	0.000*	
	coherence	0.145	0.246	0.063	0.615	**0.308**	0.000*	**0.265**	0.000*	

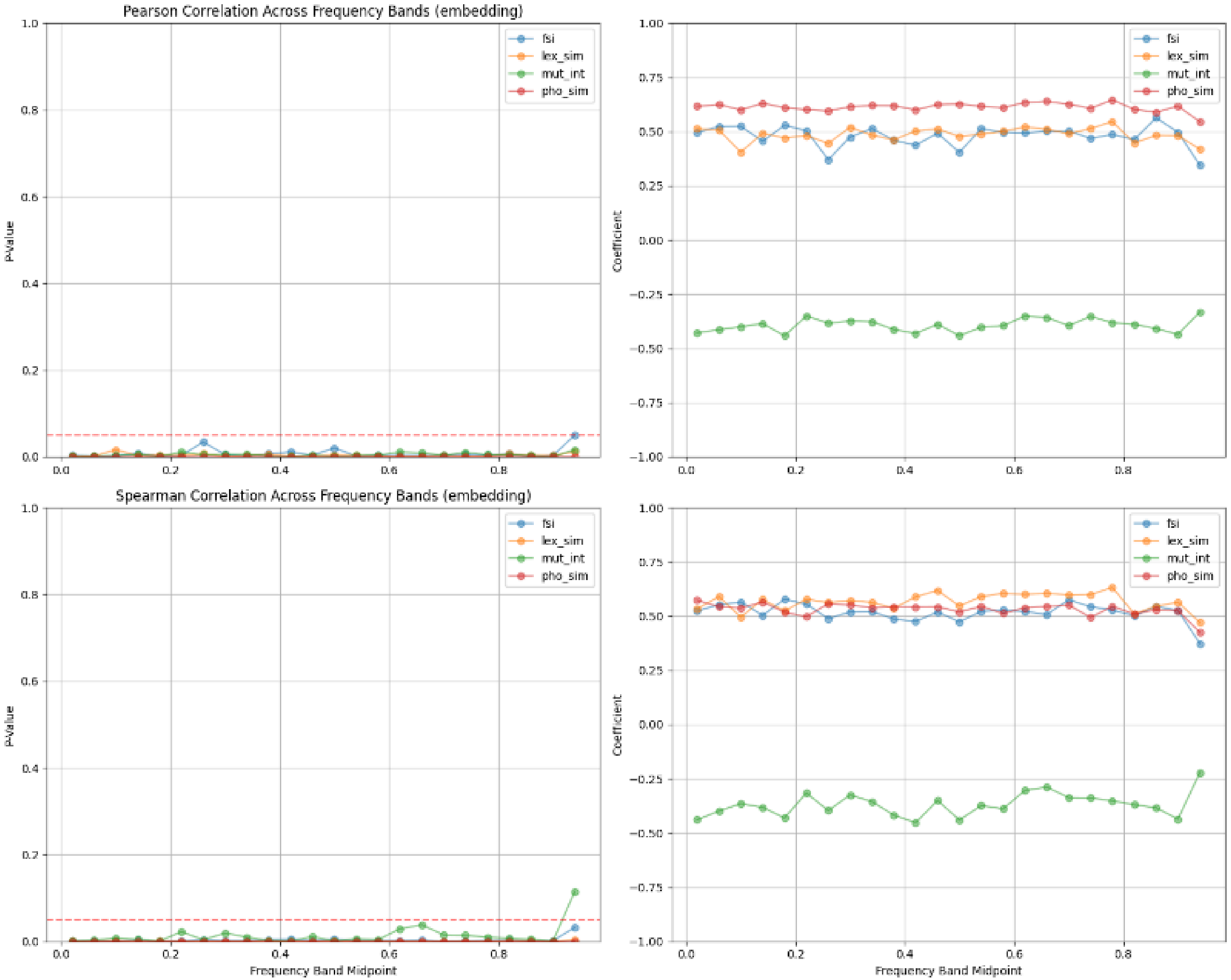
Embedding spectral metric correlations on the XNLI and Bible corpora.

Cells are highlighted when the two coefficients differ by more than 0.10; within those pairs, the larger significant coefficient is additionally shown in bold. *p < 0.05.

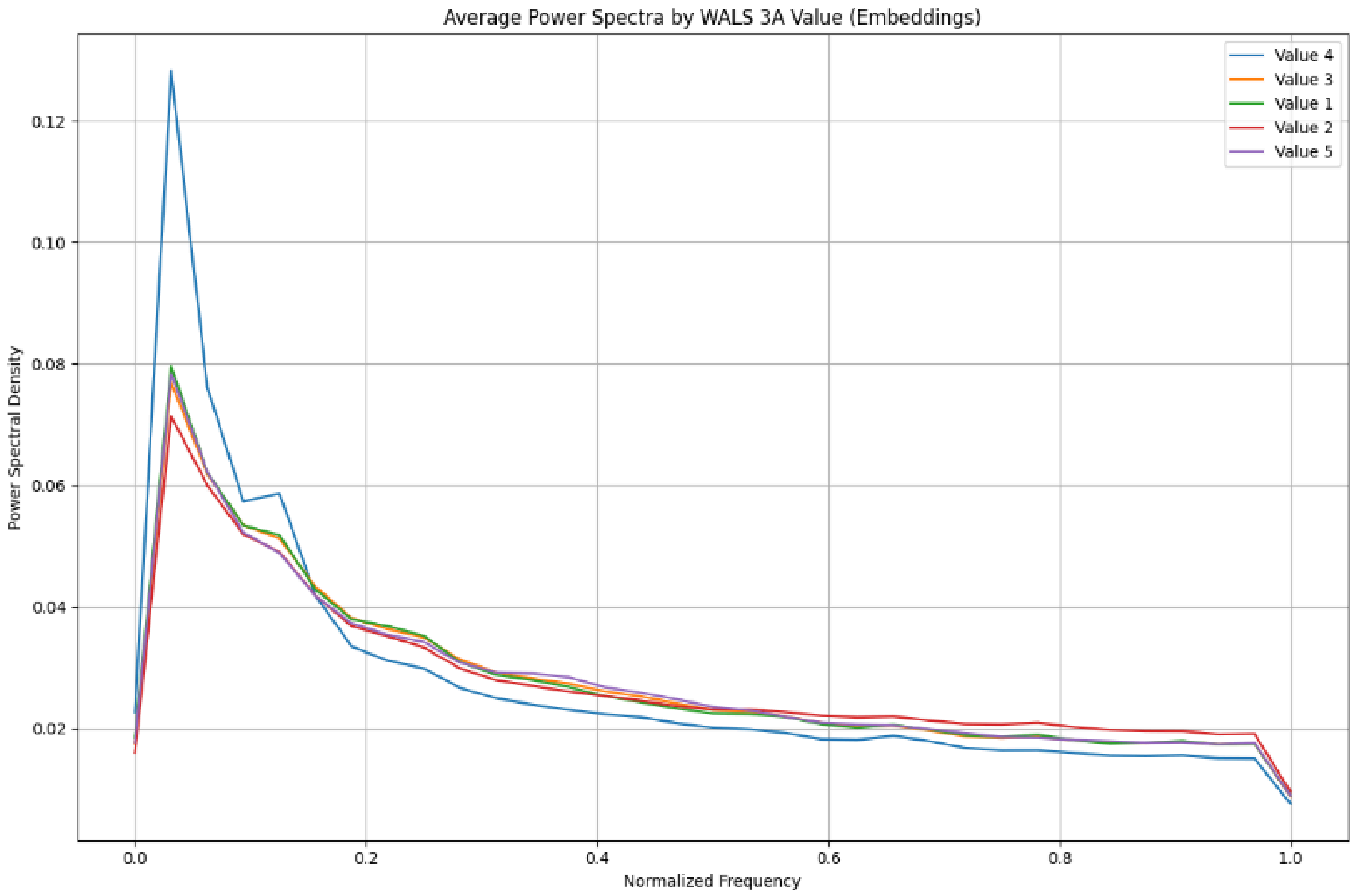
E5: mBERT Layer Investigation (Spectral Overlap)



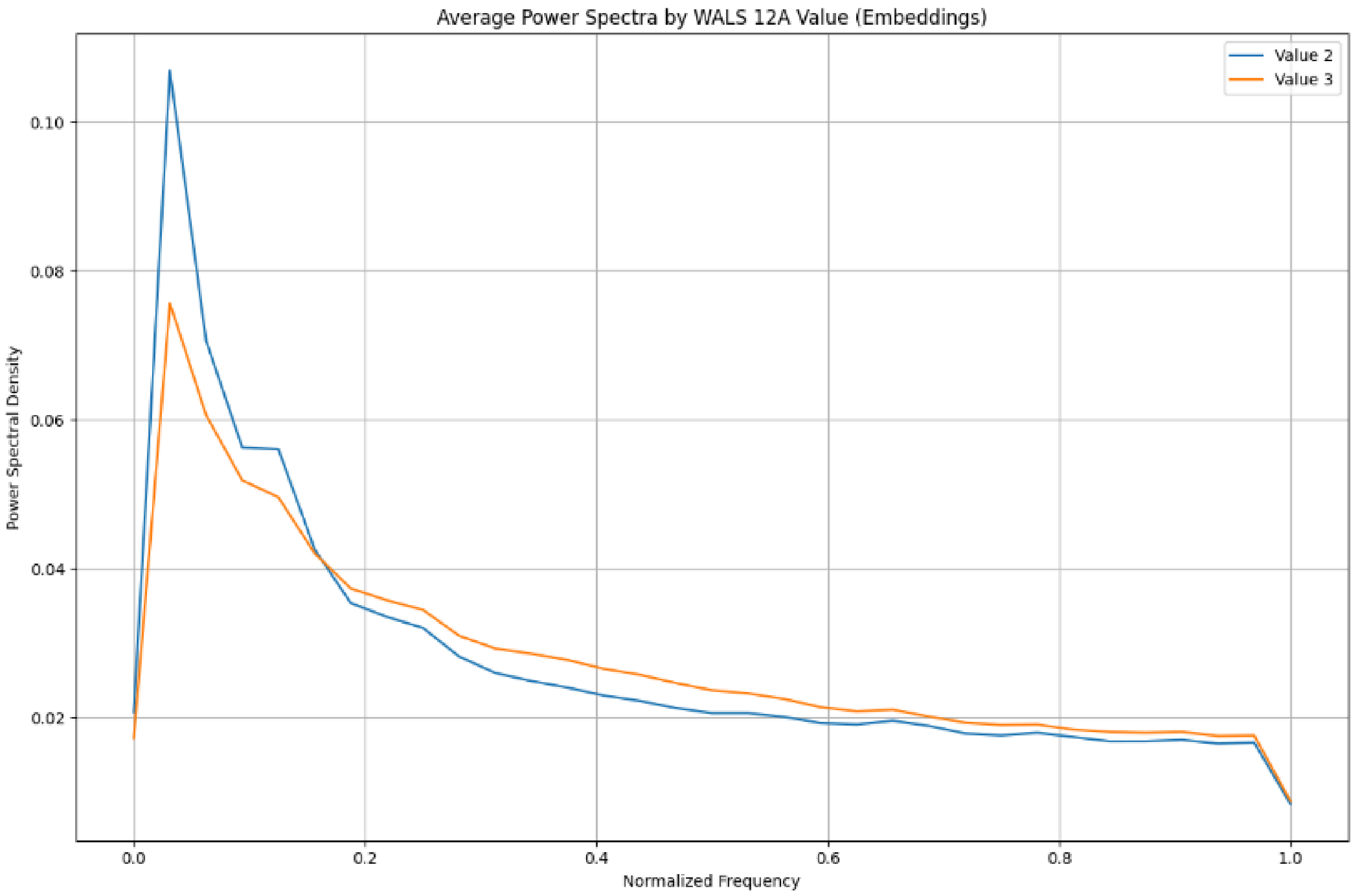
E5: mBERT Layer Investigation (Spectral KL)



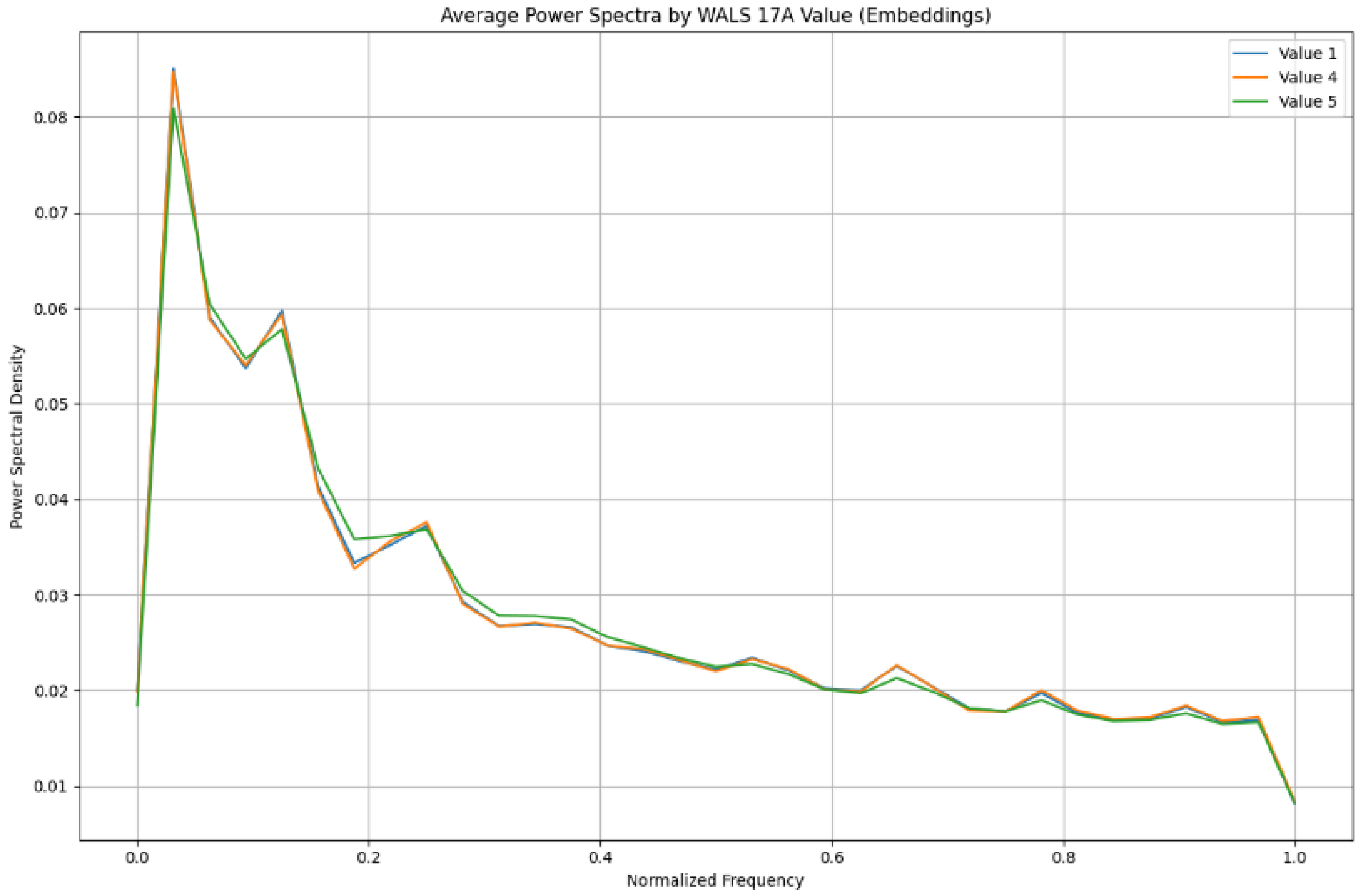
E6: Typological Modulation Effects (3A, Consonant-Vowel Ratio)



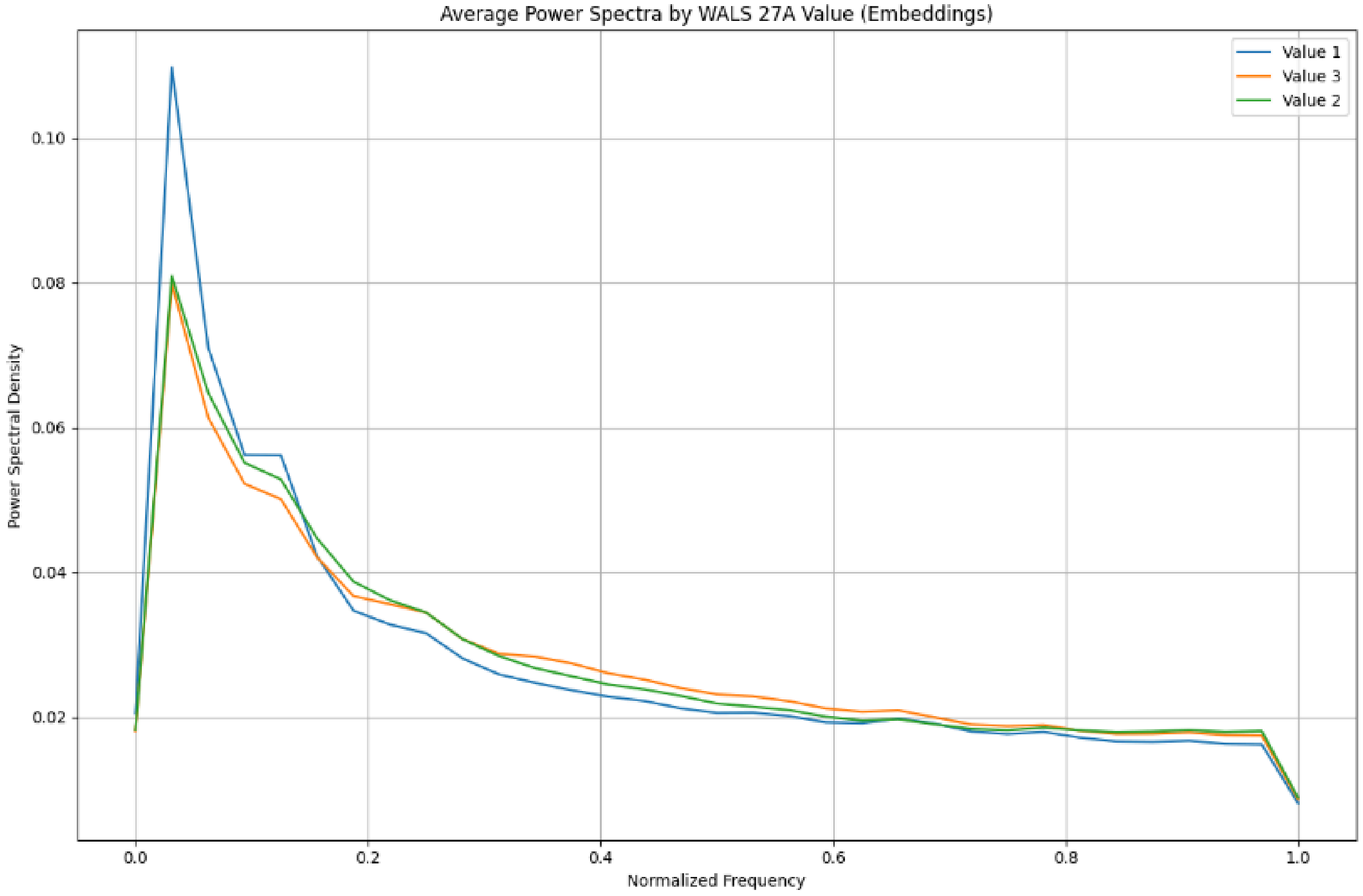
E6: Typological Modulation Effects (12A, Syllable Structure)



E6: Typological Modulation Effects (17A, Rhythm Types)

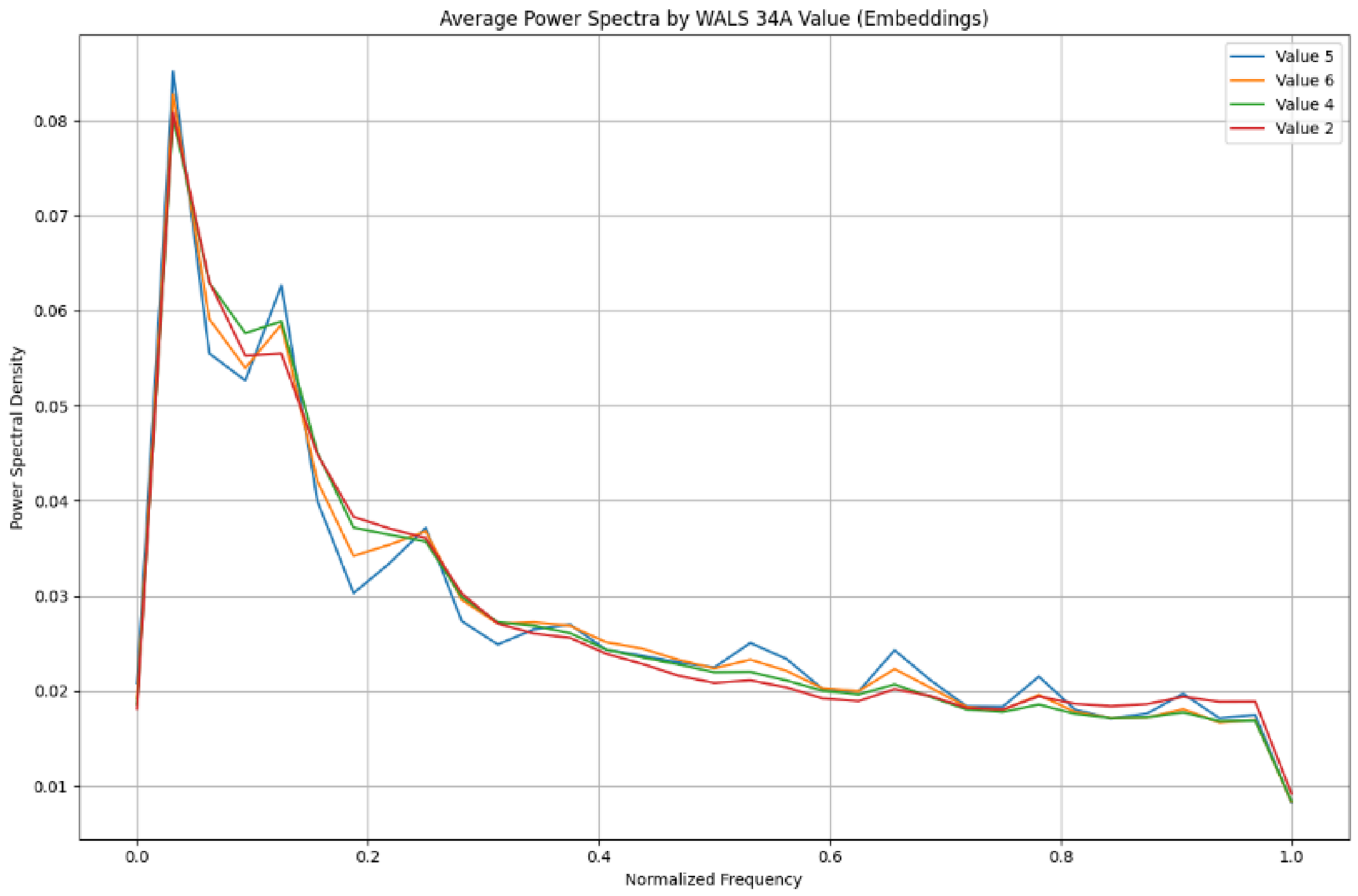


E6: Typological Modulation Effects (27A, Reduplication)

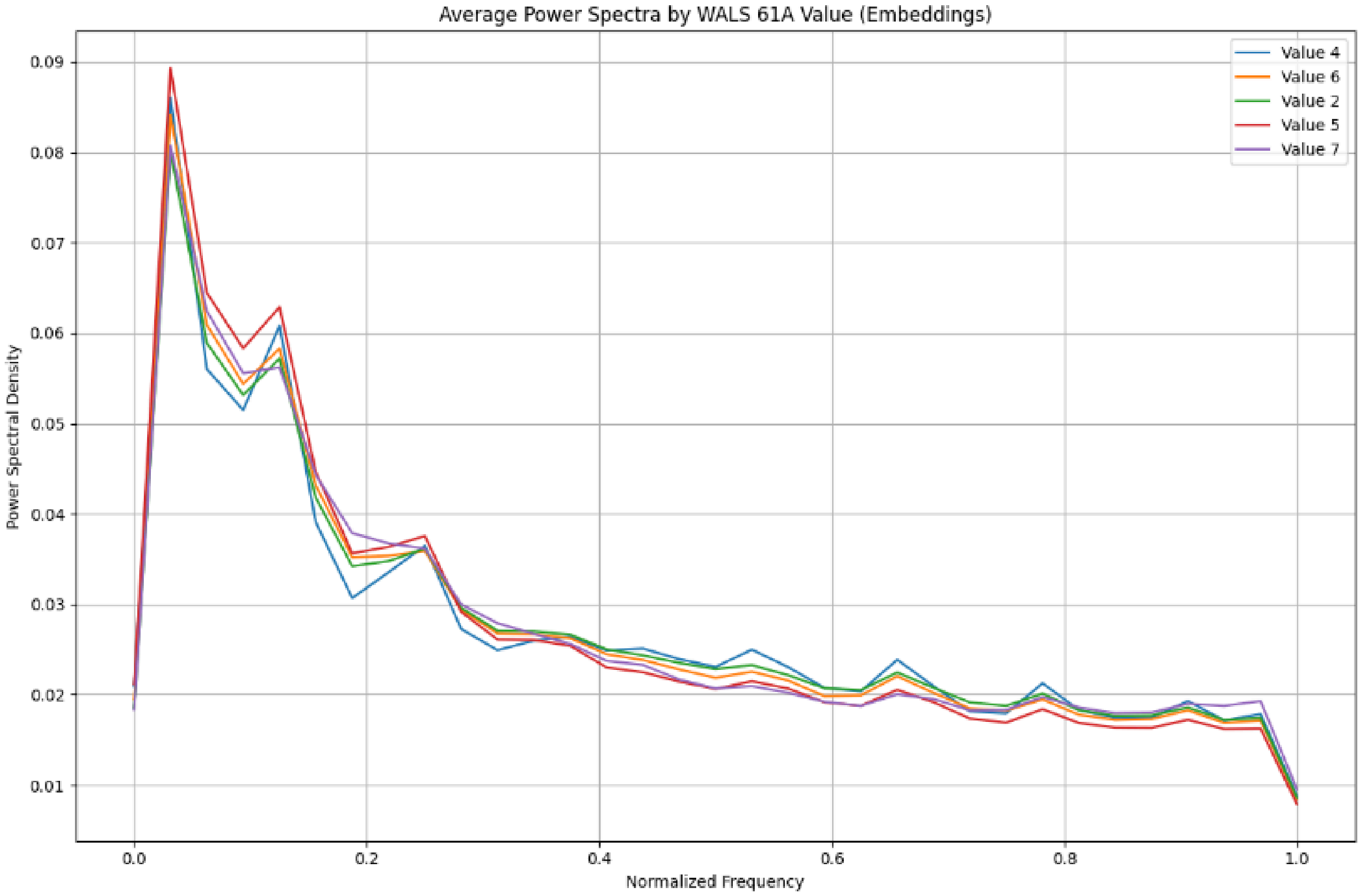


Value 1: Productive full and partial reduplication

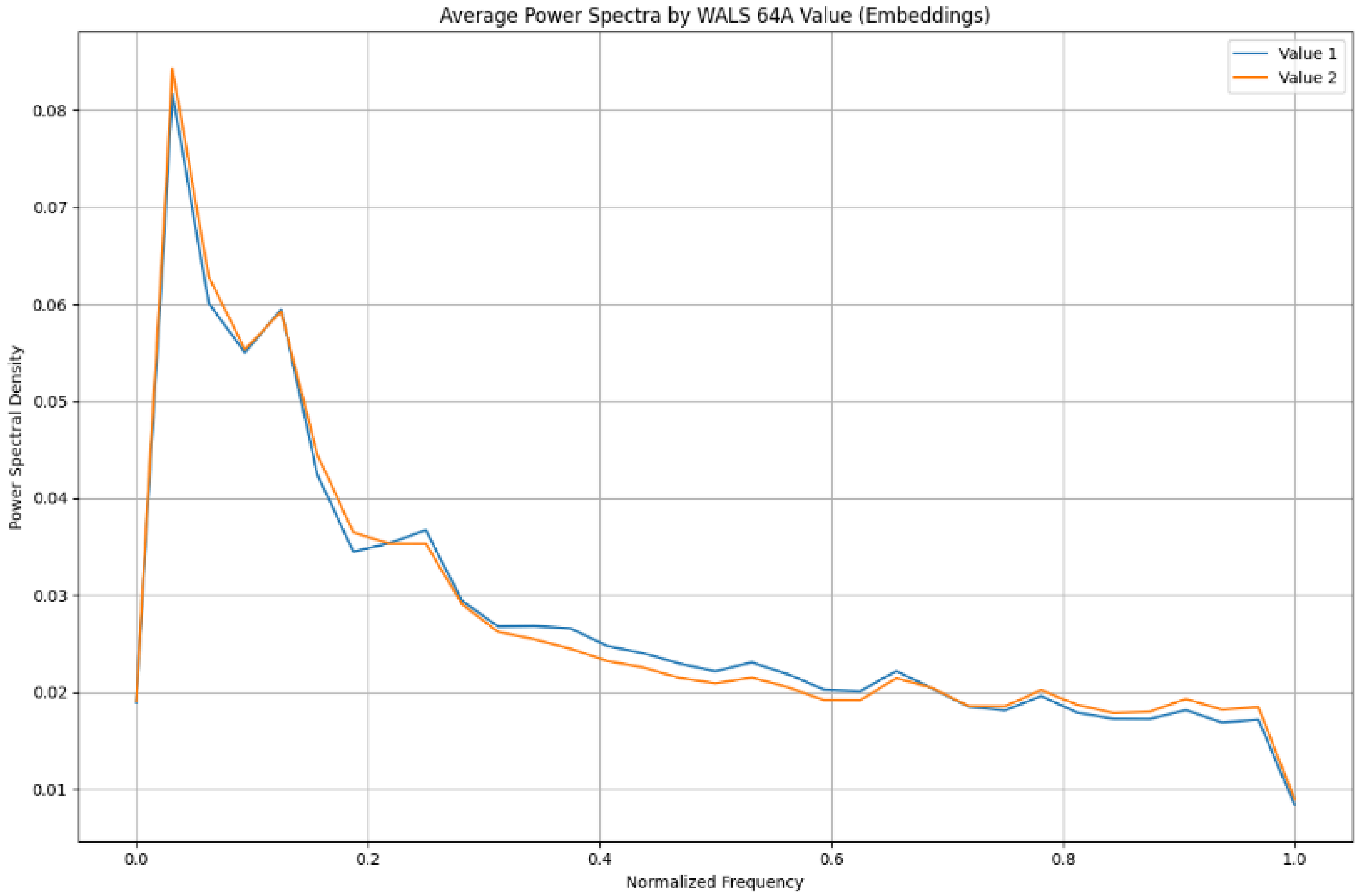
E6: Typological Modulation Effects (34A, Occurrence of Nominal Plurality)



E6: Typological Modulation Effects (61A, Adjectives without Nouns)



E6: Typological Modulation Effects (64A, Nominal and Verbal Conjunction)



E6: Typological Modulation Effects (78A, Coding of Evidentiality)